

Machine learning-driven identification of alcohol dose-dependent microbial and metabolic patterns in the human gut microbiota

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1 supplementary material

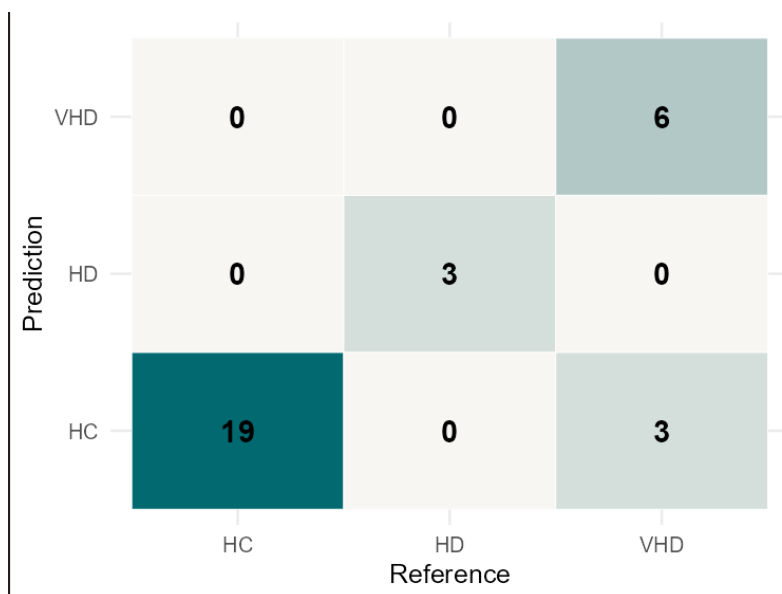


Fig. S1 Confusion matrix of the Random Forest model evaluated on Metacyc Metabolic pathways. Diagonal elements indicate correctly classified samples across the HC, HD, and VHD groups, while off-diagonal elements reflect misclassifications.

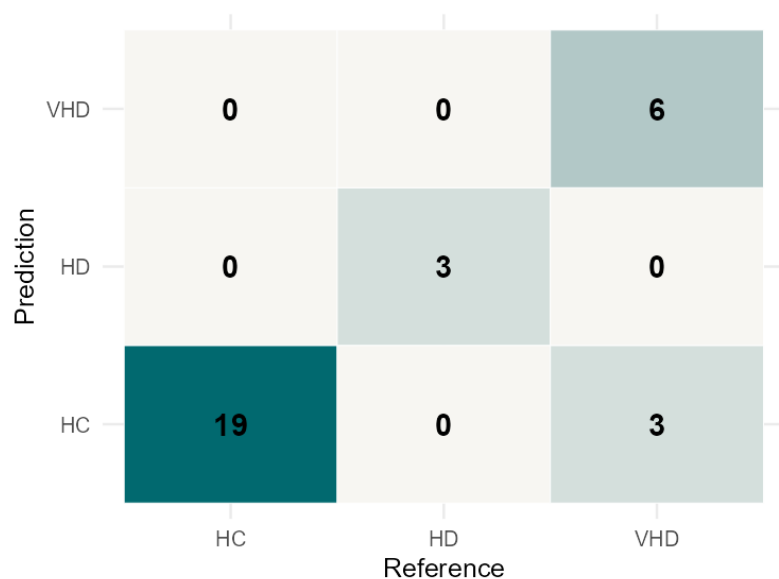


Fig. S2 Confusion matrix of the Random Forest model evaluated on Metacyc Metabolic pathways. Diagonal elements indicate correctly classified samples across the HC, HD, and VHD groups, while off-diagonal elements reflect misclassifications.

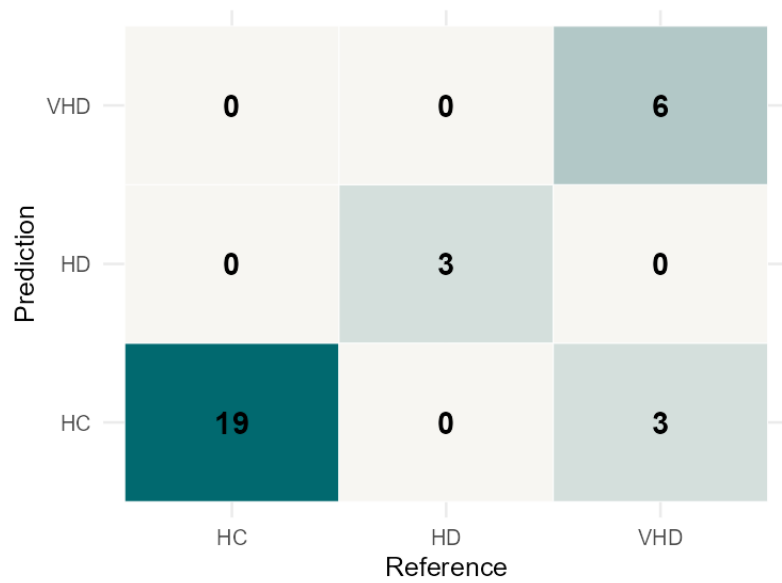


Fig. S3 Relative importance of the top 5 metabolic enzymes selected by Random Forest analysis, illustrating relative concentration trends across healthy controls (HC), Heavy Drinkers (HD), and Very Heavy Drinkers (VHD) groups.

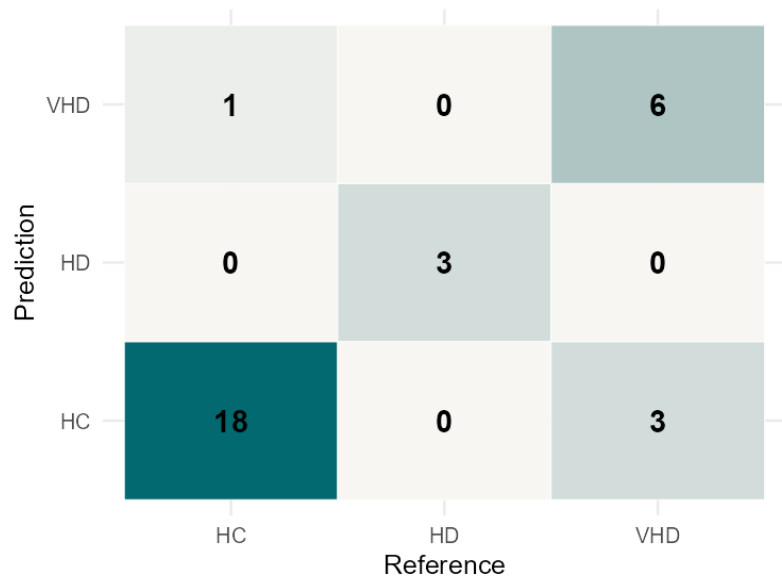


Fig. S4 Confusion matrix of the Support Vector Machine model evaluated on enzymatic functional profiles (EC numbers). Diagonal elements indicate correctly classified samples across the HC, HD, and VHD groups, while off-diagonal elements reflect misclassifications.

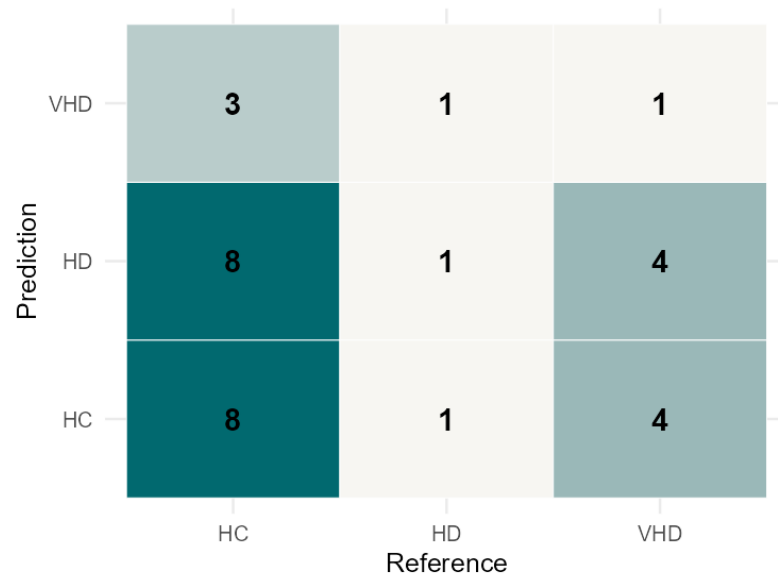


Fig. S5 Confusion matrix of the XGBoost model evaluated on Metacyc Metabolic pathways. Diagonal elements indicate correctly classified samples across the HC, HD, and VHD groups, while off-diagonal elements reflect misclassifications.

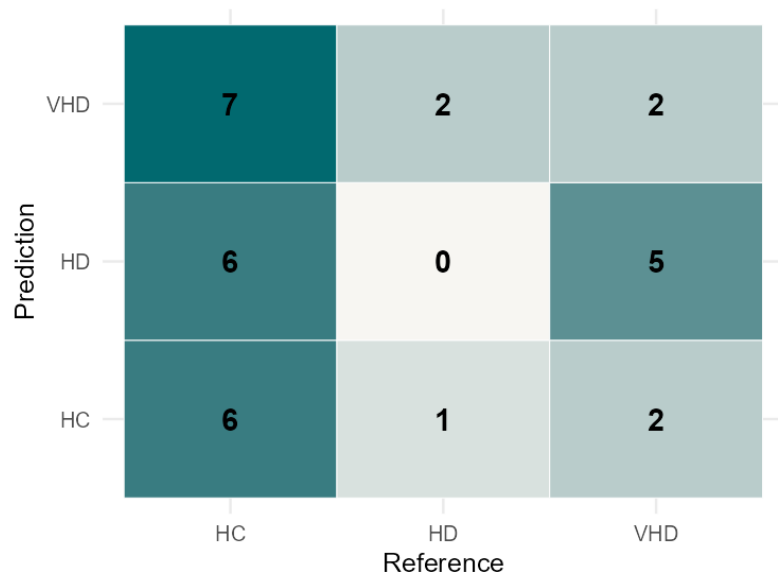


Fig. S6 Confusion matrix of the XGBoost model evaluated on enzymatic functional profiles (EC numbers). Diagonal elements indicate correctly classified samples across the HC, HD, and VHD groups, while off-diagonal elements reflect misclassifications.

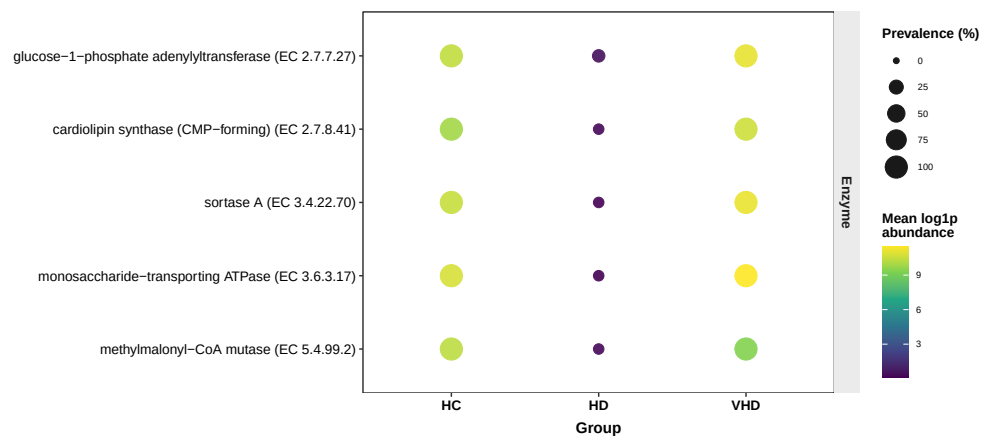


Fig. S7 Relative importance of the top 5 metabolic enzymes selected by Random Forest analysis, illustrating relative concentration trends across healthy controls (HC), Heavy Drinkers (HD), and Very Heavy Drinkers (VHD) groups.

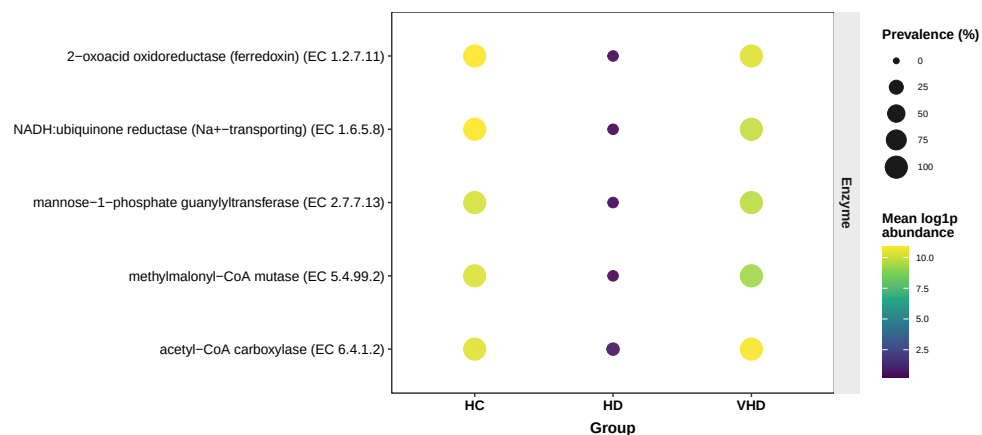


Fig. S8 Relative importance of the top 5 metabolic enzymes selected by SVM-RBF analysis, illustrating relative concentration trends across healthy controls (HC), Heavy Drinkers (HD), and Very Heavy Drinkers (VHD) groups.

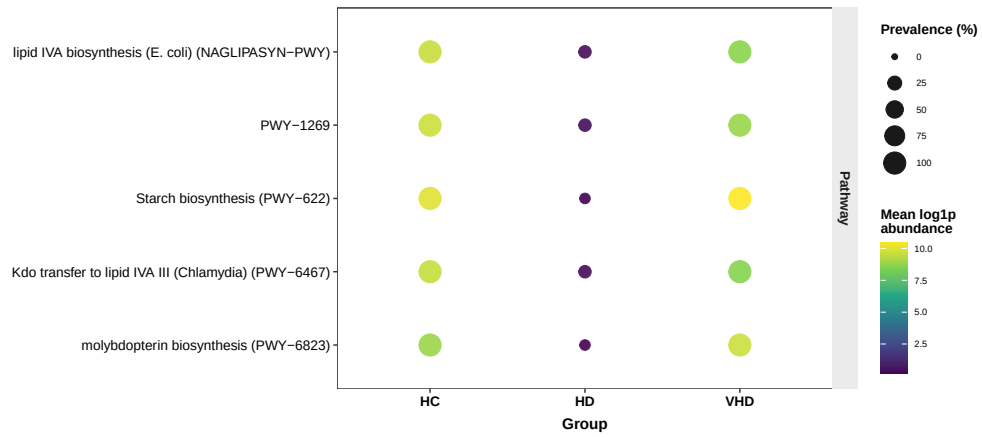


Fig. S9 Relative importance of the top 5 metabolic pathways selected by Random Forest analysis, illustrating relative concentration trends across healthy controls (HC), Heavy Drinkers (HD), and Very Heavy Drinkers (VHD) groups.

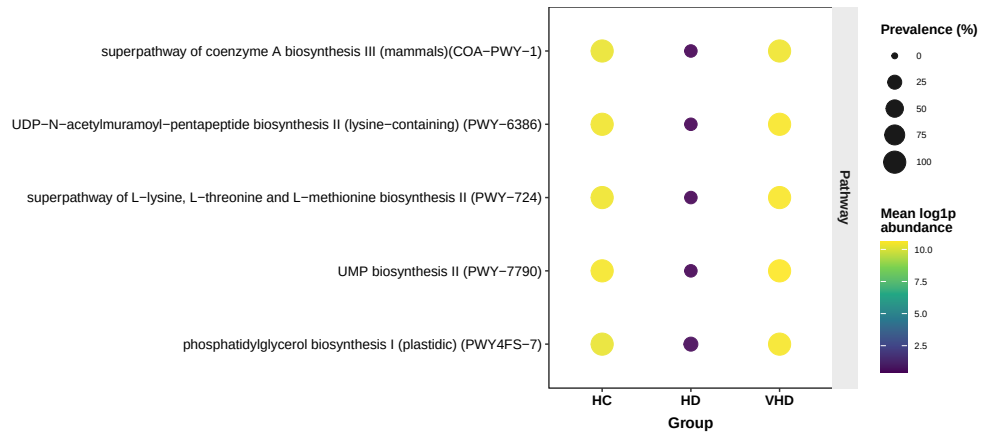


Fig. S10 Relative importance of the top 5 metabolic pathways selected by SVM-RBF analysis, illustrating relative concentration trends across healthy controls (HC), Heavy Drinkers (HD), and Very Heavy Drinkers (VHD) groups.